Enterprise Infrastructure Monitoring & Observability Platform

Architecture Design Document (Version 2.0)

Executive Summary

This document defines the production architecture for a centralized monitoring platform. Version 1 validated the technology stack but exposed architectural limitations. Version 2 standardizes networking, monitoring, alerting and operations to provide a scalable enterprise foundation.

Business Objectives

Provide centralized monitoring, proactive alerting, centralized logging, hardware visibility, application monitoring, faster troubleshooting, reduced downtime and future scalability.

Current Environment

DigitalOcean monitoring server, Proxmox, Dell servers, pfSense, Cisco & HP switches, Node.js, Python, PHP, MySQL, PostgreSQL, Redis and hosted websites.

Proof of Concept Review

Ubuntu, Docker, Prometheus, Grafana, Loki, Alertmanager, Blackbox Exporter and SNMP Exporter were deployed successfully. During onboarding of the first monitored server, networking and configuration issues demonstrated that the architecture should be redesigned before expanding.

Issues Found

Mixed Docker networking, Docker DNS limitations, Grafana configuration override, increasing compose complexity and lack of modular structure.

Rebuild Strategy

Create a modular production platform with consistent networking, structured configuration, provisioning, exporters, documentation and an independent Ubuntu watchdog.

Alerting

Alertmanager remains the primary notification engine. A native Ubuntu Watchdog monitors Docker and the monitoring platform itself. External uptime monitoring detects total server failure. This avoids duplicate alerts and circular dependencies.

Implementation Roadmap

Rebuild core platform, monitor monitor01, Docker monitoring, Proxmox, physical servers, network devices, databases, applications, centralized logging, dashboards and operations.

Target Architecture

```
External Uptime Monitor
          |
Ubuntu Watchdog (systemd)
          |
Ubuntu Monitoring Server
          |
Docker
 |
 +-- Grafana
 +-- Prometheus
 +-- Alertmanager
 +-- Loki
 +-- Grafana Alloy
 +-- Node Exporter
 +-- cAdvisor
 +-- Process Exporter
 +-- SNMP Exporter
 +-- Blackbox Exporter
          |
On-Prem Infrastructure
(Proxmox, Servers, Switches,
Databases, Applications)
```